

Ethan Sequeira

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Work Experience

Martinrea International | Software Engineering Lead Sep 2025 – Present | Vaughan, ON

- Leading a team to develop and ship a distributed resistance spot welding optimization system, driving product strategy and roadmap for a planned North American plant-wide deployment in 2026
- Built a plant-server data broker supporting dynamic app registration, multi-protocol file transfer (HTTP/tus), and bidirectional job lifecycle management across edge devices
- Implemented full observability infrastructure using OpenTelemetry and the Grafana LGTM stack
- Drove department-wide standardization on Linear with structured deployment-issue tracking; built an AI agent that automatically extracts operational insights from issue history, replacing manual report compilation
- Presenting technical strategy and product updates to the CEO and executive team biweekly

Martinrea International | Software Engineer & Technical Lead Apr 2024 – Aug 2025 | Vaughan, ON

- Led technical architecture and development of distributed resistance spot welding optimization system, collaborating with cross-functional teams and external manufacturing partners
- Developed high-performance event-driven C++ application that processes real-time weld data and executes optimization jobs, achieving 30x performance improvement while significantly decreasing plant network traffic
- Architected distributed system with intelligent job broker that consolidated hardware requirements, enabling one edge device to service multiple manufacturing lines versus the previous multi-device-per-line setup
- Drove development of process parameter optimization algorithms in collaboration with weld engineers, while managing full-stack monitoring application and product requirements

Martinrea International | Software Engineer (Co-op) Summers 2021-2023 | Vaughan, ON

- Co-led an R&D team building hardware and software solutions for manufacturing vision applications in Python and C++
- Created a Python library that abstracted complex computer vision workflows and reduced development time. Extended this with a configurable data augmentation framework with spatial and pixel-level transformations for ML pipelines
- Worked across multiple AMR subsystems for factory inventory management using Java, Python, C++, and CUDA.
- Achieved a 2500% performance boost in AMR image processing pipelines and enhanced SLAM system with save/load functionality and ArUco-based camera calibration.

Projects

Asobi | Founding Engineer May 2025 - Present | Toronto, ON

- Building a multi-tenant SaaS platform for sports training businesses. Actively serving 500+ customers and processing \$150K+ in transactions with a single tenant
- Shipping full-stack features across scheduling, payments, booking flows, and customer management using Django, PostgreSQL, and Next.js, deployed on AWS ECS. Built production observability stack with Grafana Cloud

VieCloud | Capstone Project Sep 2023 - Apr 2024 | Toronto, ON

- Developed a real-time cloud application with AWS to stream and process 64-channel EEG data, building the data pipeline from hardware acquisition through automated signal processing and analysis
- Built a full-stack web app for Vielight, a biotechnology company, creating an automated closed-loop system that visualizes real-time brain wave data and dynamically adjusts photobiomodulation device parameters

Research

McMaster University | Research Assistant Aug 2023 - Jun 2024 | Hamilton, ON

- Under Dr. Matthew Giamou's supervision, published a paper on optimal beacon placement for range-based localization systems, developing an information-theoretic method with provable performance guarantees
- Built physics simulator and ran 2D/3D experiments comparing the greedy algorithm against multiple benchmark heuristics in python

Education

McMaster University Hamilton, ON

B. Eng. Software and Biomedical Engineering (Co-op), Class of 2024

Technical Languages & DBs Python • C++ • Java • Shell scripting • SQL • Postgres • MongoDB • Redis

Tools Django • FastAPI • AWS • Docker • Git • RabbitMQ • Celery • Linux • OpenCV • CUDA • OpenTelemetry • Grafana

Certifications NVIDIA Generative AI Multimodal